

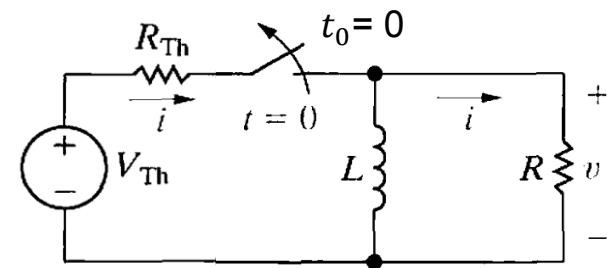
ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Response of Inductor, Capacitor in Electrical Circuits

Recap: Response in circuit with DC sources and switches



$$v(t > 0) = I_0 R e^{-\frac{R}{L}t}$$

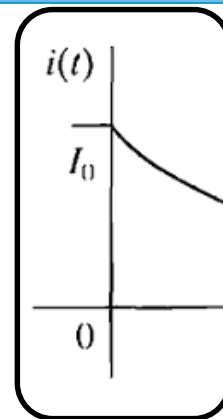
$$P(t > 0) = I_0^2 R e^{-2\frac{R}{L}t}$$

$$\text{At } t = 0, \quad L \frac{di}{dt} + Ri = 0$$

$$\Rightarrow \int_{i(t_0)}^{i(t)} \frac{di}{i} = - \int_{t_0}^t \frac{R}{L} dt$$

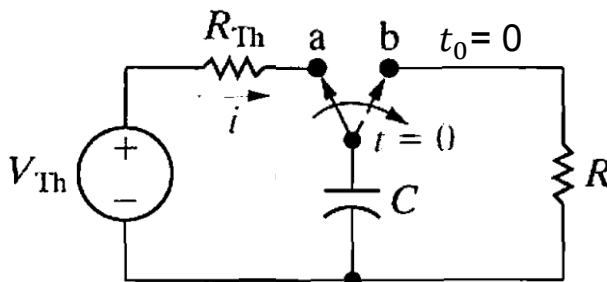
$$\Rightarrow i(t) = i(t_0) e^{-\frac{R}{L}t}$$

$$\text{Here, } i(t_0) = I_0 = \frac{V_{th}}{R_{th}}$$



Transient Response

After a long time, steady state response



$$i(t > 0) = \frac{V_0}{R} e^{-\frac{t}{RC}}$$

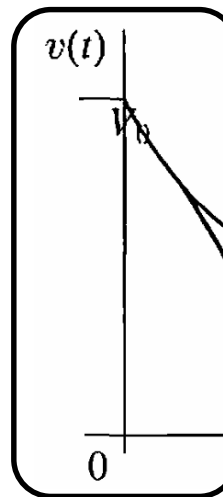
$$P(t > 0) = \frac{V_0^2}{R} e^{-\frac{2t}{RC}} \Rightarrow E(t > 0) = \frac{CV_0^2}{2} (1 - e^{-\frac{2t}{RC}})$$

$$\text{At } t = 0, \quad C \frac{dv}{dt} + \frac{v}{R} = 0$$

$$\Rightarrow \int_{v(t_0)}^{v(t)} \frac{dv}{v} = - \int_{t_0}^t \frac{1}{RC} dt$$

$$\Rightarrow v(t) = v(t_0) e^{-\frac{t}{RC}}$$

$$\text{Here, } v(t_0) = V_0 = V_{th}$$



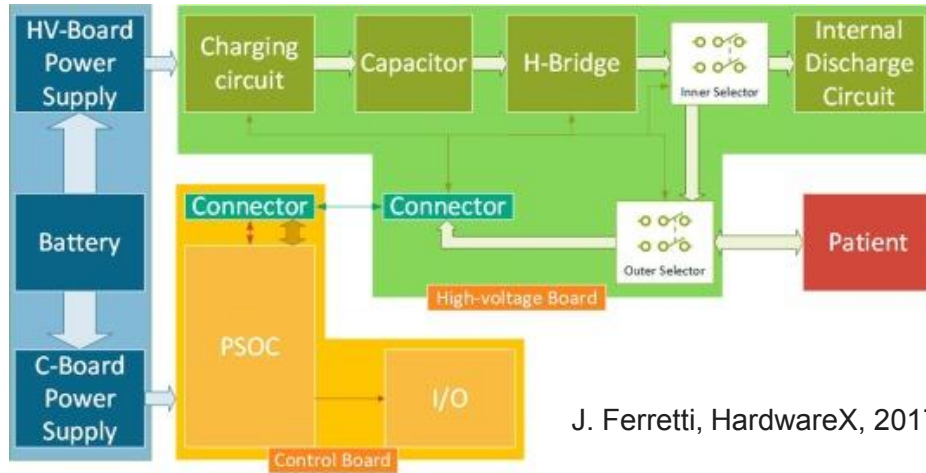
Transient Response

$$v(t) = V_0 e^{-t/\tau}$$

$$v(t) = V_0 - \frac{V_0}{\tau} t$$

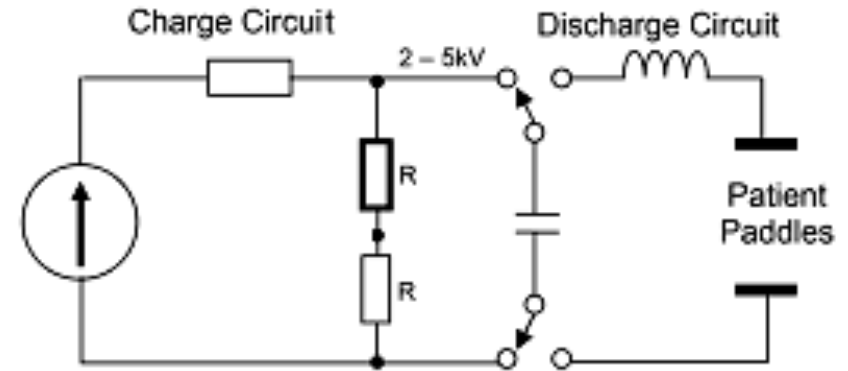
steady state response

Applications where inductor/capacitor's natural response is used



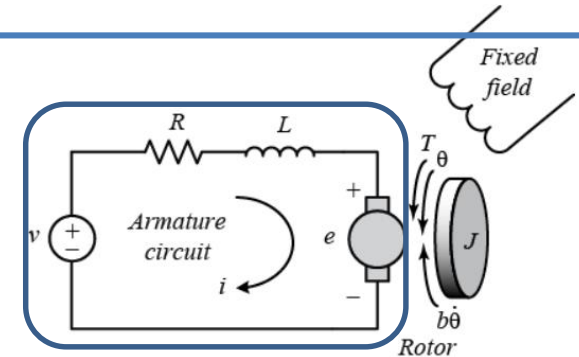
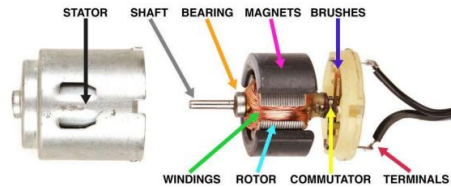
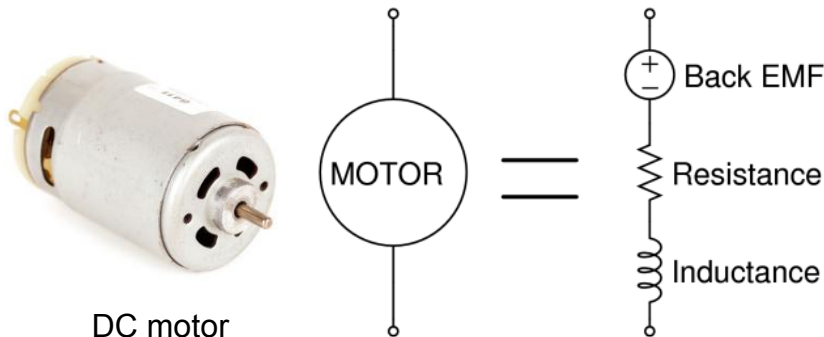
J. Ferretti, HardwareX, 2017.

Block diagram of an Automated External Defibrillator (AED)



A simple circuit representation

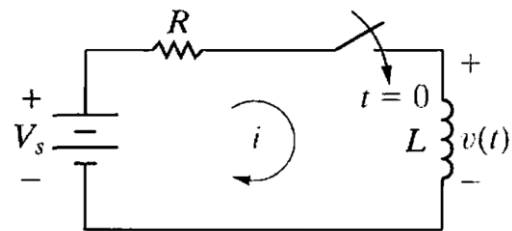
Components, AC Source: The wall outlet, Switch, R : The wire resistance, L : The field windings.



Electrical Equivalent circuit

What could happen when it is switched off?

Recap: Response in circuit with DC sources and switches



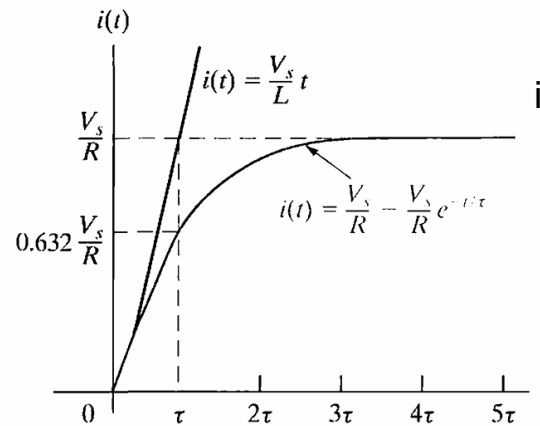
Here, we assume $i(t_0) = I_0$

$$\text{At } t = 0, \quad L \frac{di}{dt} + Ri = V_s$$

$$\Rightarrow \int_{i(t_0)}^{i(t)} \frac{di}{i - (V_s/R)} = - \int_{t_0}^t \frac{R}{L} dt$$

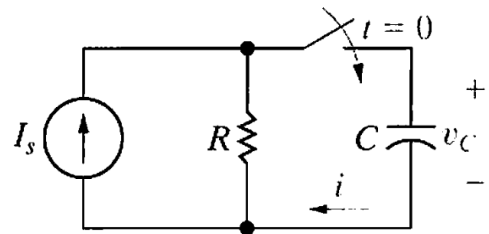
$$\Rightarrow i(t) = \frac{V_s}{R} + (I_0 - \frac{V_s}{R})e^{-\frac{R}{L}t}$$

$$\text{if } I_0 = 0, \quad i(t) = \frac{V_s}{R} - \frac{V_s}{R}e^{-\frac{R}{L}t}$$



$$v(t) = V_s - i(t)R$$

$$\text{if } I_0 = 0, \quad v(t) = V_s e^{-\frac{R}{L}t}$$



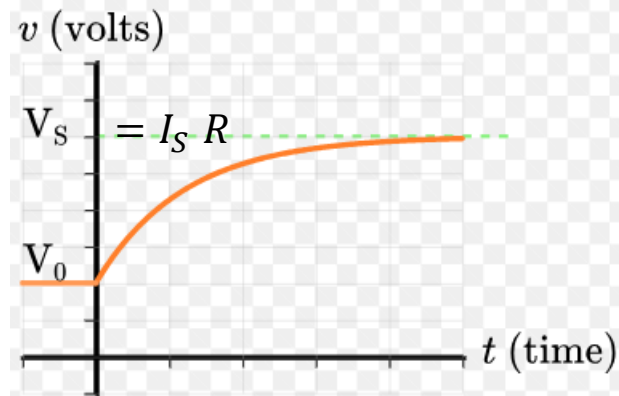
Here, we assume $v(t_0) = V_0$

$$\text{At } t = 0, \quad C \frac{dv_c}{dt} + \frac{v_c}{R} = I_s$$

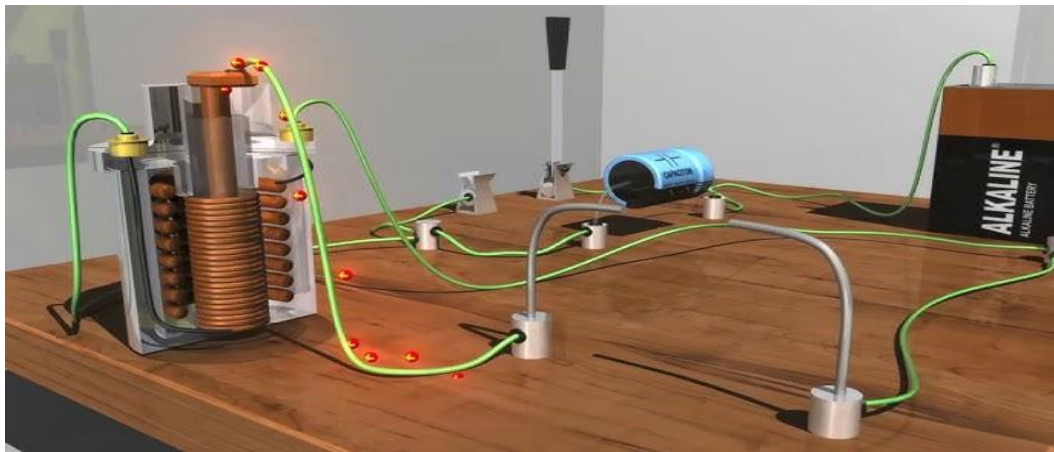
$$\Rightarrow \int_{v(t_0)}^{v(t)} \frac{dv_c}{v_c - I_s R} = - \int_{t_0}^t \frac{1}{RC} dt$$

$$\Rightarrow v_c(t) = I_s R + (V_0 - I_s R)e^{-\frac{t}{RC}}$$

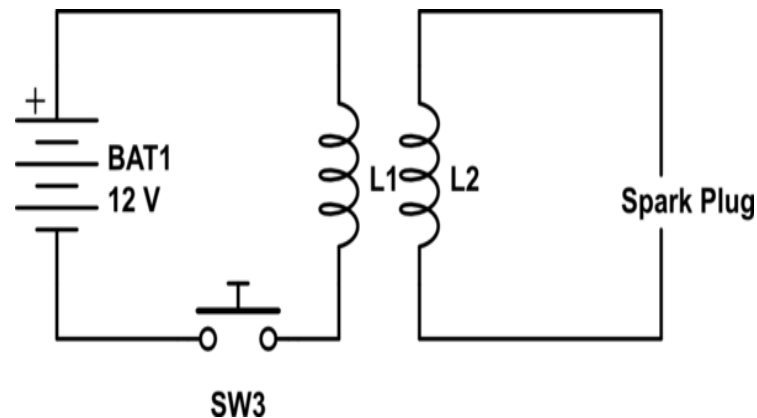
$$i(t) = I_s - \frac{v_c(t)}{R} = (I_s - \frac{V_0}{R})e^{-\frac{t}{RC}}$$



Applications where L & C's step response is often encountered



[Creation of Spark in Ignition coil](#) (video)



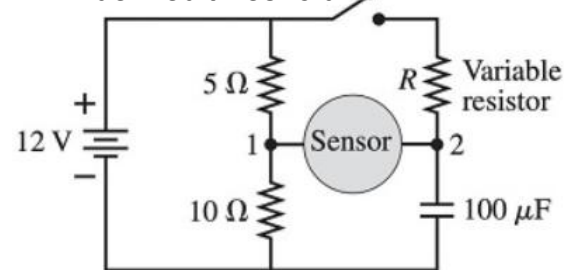
Representative Electrical Circuit where Primary coil (L1) induces the magnetic field in secondary coil (L2) when SW3 is closed.



Intermittent Windshield Wipers

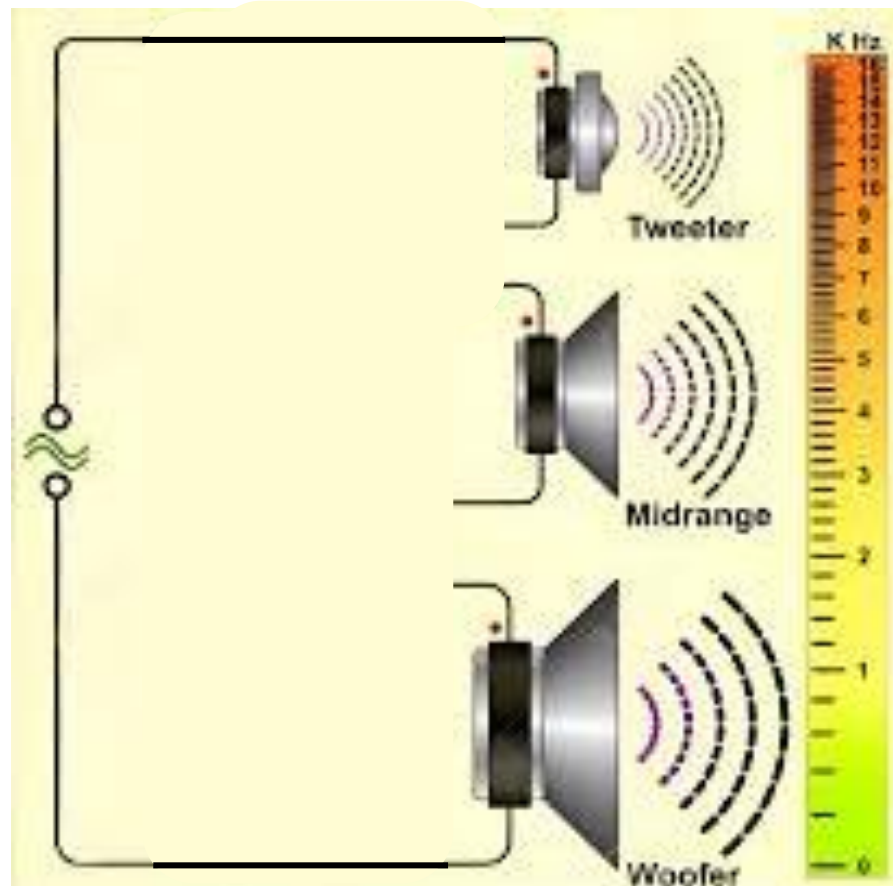
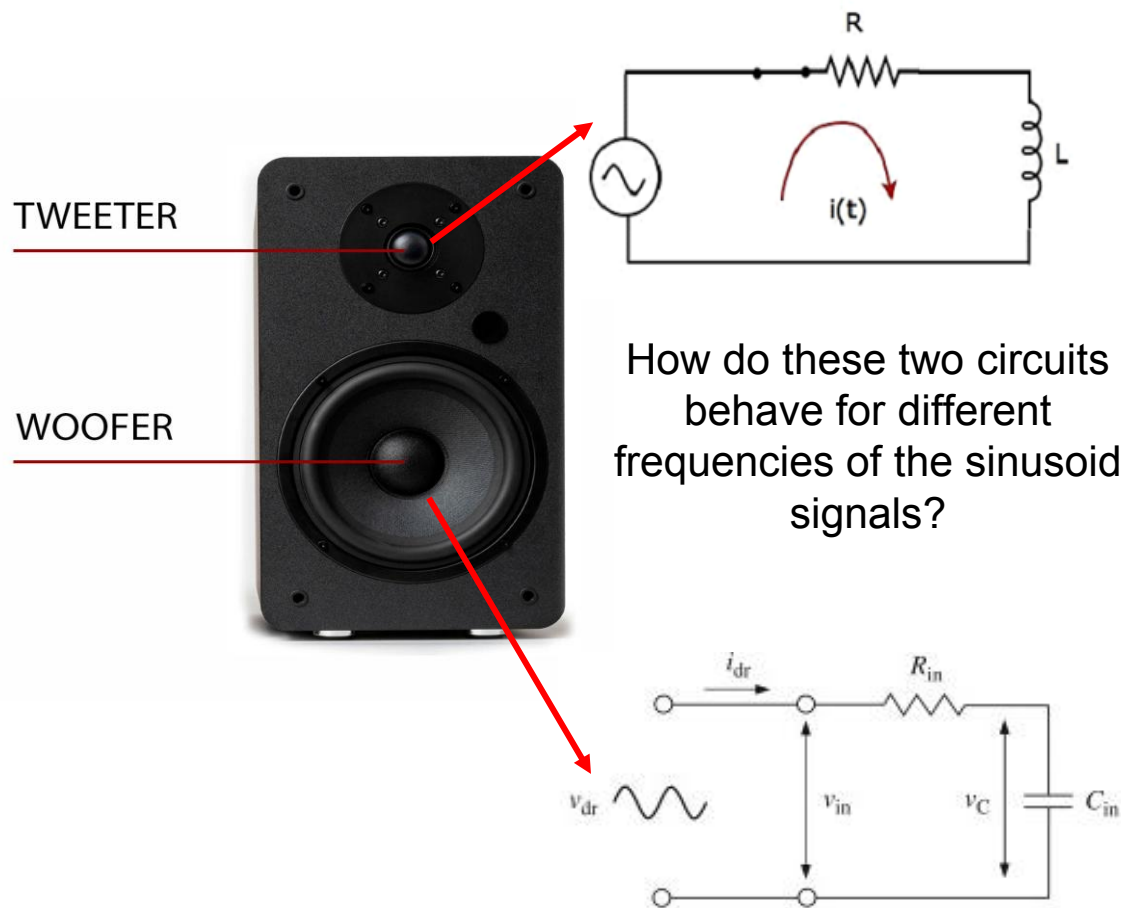


Motor starts once the capacitor voltage reaches a pre-defined threshold

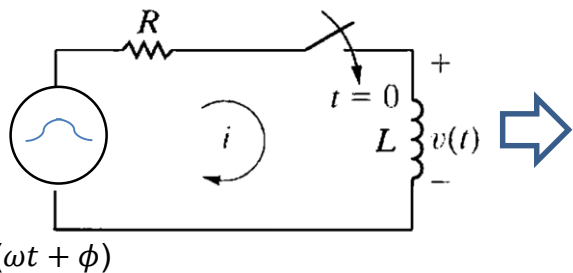


Representative Electrical Circuit where variable resistor is set by knob for creating different delay

Applications where L & C's sinusoid response is often encountered



Response in circuit with AC (sinusoid) sources and switches



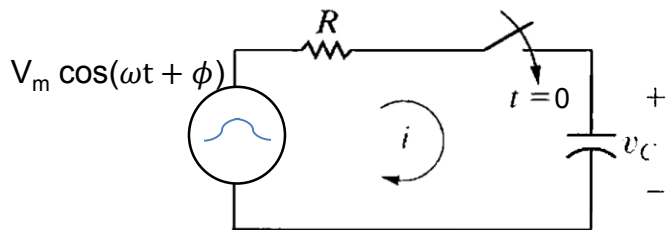
$$V_s = V_m \cos(\omega t + \phi) \quad \text{At } t = 0, \quad L \frac{di}{dt} + Ri = V_s = V_m \cos(\omega t + \phi)$$

$$\Rightarrow \int_{i(t_0)}^{i(t)} \frac{di}{i - (v_s/R)} = - \int_{t_0}^t \frac{R}{L} dt$$

$$i = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta)$$

Here, $\theta = \tan^{-1} \frac{\omega L}{R}$

Can we break down this circuit analysis into two cases:
1) Circuit subjected to no source (i.e. natural response),
2) The forced response from sinusoidal source?



$$\text{At } t = 0, \quad RC \frac{dv_c}{dt} + v_c(t) = v_s = V_m \cos(\omega t + \phi)$$

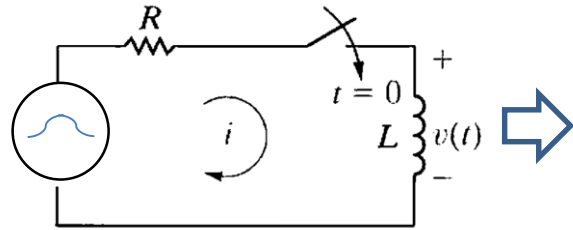
$$\text{Ansatz } v_c(t) = A \cos(\omega t + \phi) + B \exp\left[\frac{-t}{RC}\right]$$

$$i(t) = C \frac{dv_c(t)}{dt} = \frac{v_o C \omega}{\cos \phi - \omega RC \sin \phi} \cos\left(\omega t + \phi + \frac{\pi}{2}\right)$$

Can we break down this circuit analysis into two cases:
1) Circuit subjected to no source (i.e. natural response),
2) The forced response from sinusoidal source?

Here, A and B can be obtained from the differential equation.

Response in time domain



$$V_m \cos(\omega t + \phi)$$

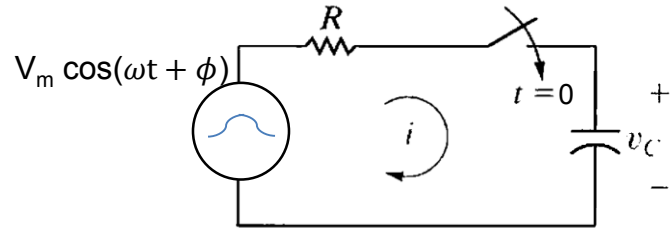
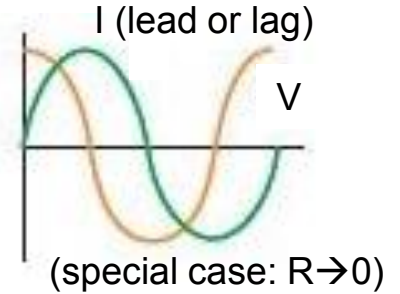
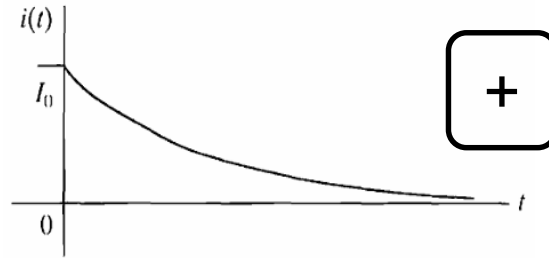
Transient response needs to be accounted for as we know that circuit with inductor/capacitor can not upstart instantly when the switch is closed at $t=0$.

$$V_s = V_m \cos(\omega t + \phi)$$

Here,

$$\theta = \tan^{-1} \frac{\omega L}{R}$$

$$i = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta)$$



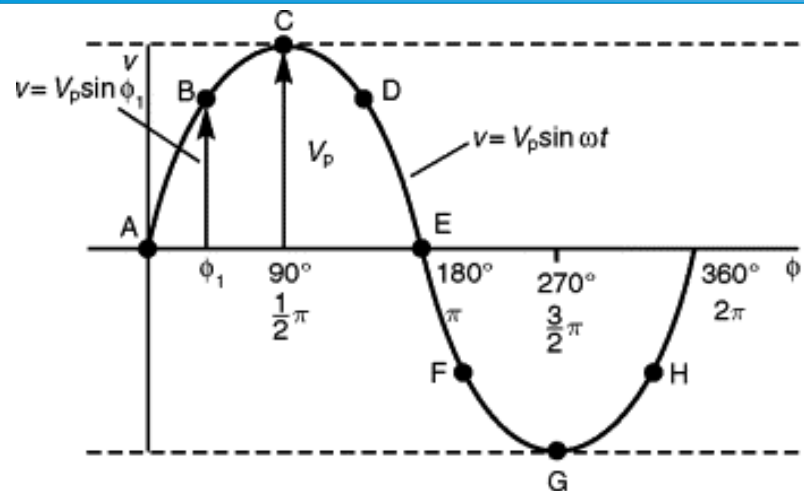
$$i(t) = C \frac{dv_c(t)}{dt} = \frac{v_o C \omega}{\cos \phi - \omega R C \sin \phi} \cos\left(\omega t + \phi + \frac{\pi}{2}\right)$$

Similarly, we will see transient term + sinusoidal term in the sinusoidal response of the R-C circuit

Transient response needs to be accounted for as we know that circuit with inductor/capacitor can not upstart instantly when the switch is closed at $t=0$.

Is there a better way to visualize the information of phase or frequency of sinusoid signals in respective domains?

Concept of Phasor Diagrams



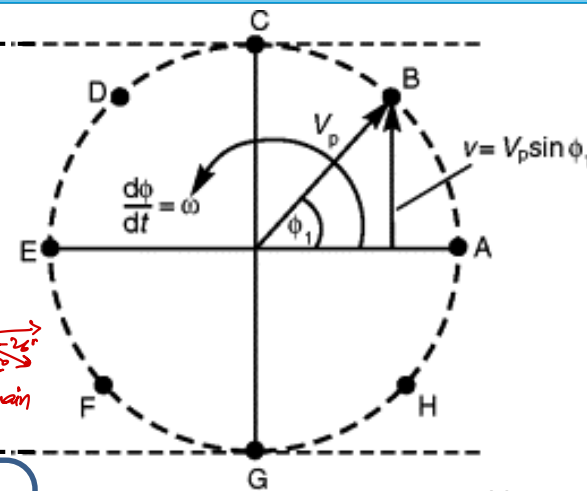
While the voltage is spinning 60 times a second (e.g. 60 Hz) is like trying to read a label on a spinning fan.

What if we could use a strobe light to 'freeze' the spinning motion and just look at the relative angles?"

i.e. $100 \cos(\omega t - 26^\circ) \rightarrow 100 \angle -26^\circ$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

vector in the phasor domain

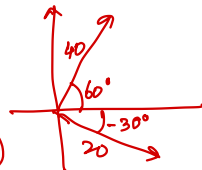


Q. Add $20 \cos(\omega t - 30^\circ)$ and $40 \cos(\omega t + 60^\circ)$

$$= 20 \angle -30^\circ + 40 \angle 60^\circ$$

$$= 20 (\cos(-30^\circ) + j \sin(-30^\circ)) + 40 (\cos(60^\circ) + j \sin(60^\circ))$$

$$1 \angle \phi^\circ = 1 \cdot e^{j\phi}$$

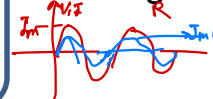


For resistor R, if $i = I_m \cos(\omega t + \phi)$

$$\Rightarrow v = I_m R \cos(\omega t + \phi)$$

In phasor domain, $\mathbf{V} = \boxed{I_m R} \angle \phi = RI$

Rotating Vector: $1/\phi = 1e^{j\phi}$



For inductor L, if $i = I_m \cos(\omega t + \phi)$

$$\Rightarrow v = -\omega L I_m \cos(\omega t + \phi - 90^\circ)$$

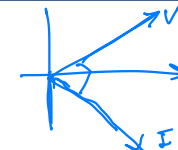
In phasor domain, $\mathbf{V} = -\omega L I_m e^{j(\phi - 90^\circ)} = j\omega L I_m e^{j\phi} = \underline{j\omega L I}$



For capacitor C, if $v = V_m \cos(\omega t + \phi)$

$$\Rightarrow i = -\omega C V_m \cos(\omega t + \phi - 90^\circ)$$

In phasor domain, $\mathbf{I} = j\omega C \mathbf{V} \Rightarrow \mathbf{V} = (1/j\omega C) \mathbf{I}$



Impedance and Reactance

For resistor R , if $i = I_m \cos(\omega t + \phi)$
 $\Rightarrow v = I_m R \cos(\omega t + \phi)$
 In phasor domain, $\underline{V} = I_m R / \phi = RI$

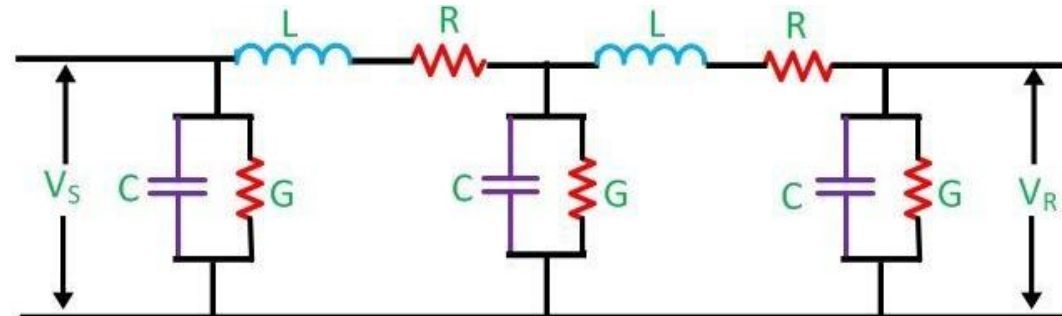
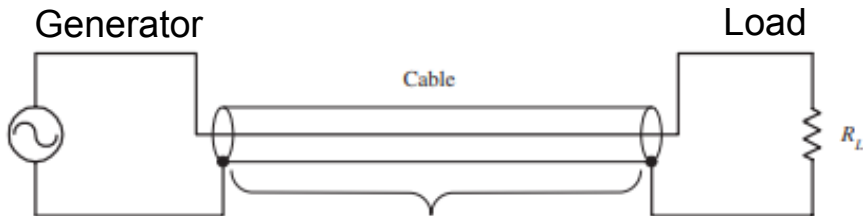
For inductor L , if $i = I_m \cos(\omega t + \phi)$
 $\Rightarrow v = -\omega L I_m \cos(\omega t + \phi - 90^\circ)$
 In phasor domain, $\underline{V} = -\omega L I_m e^{j(\phi - 90^\circ)} = j\omega L I_m e^{j\phi} = j\omega LI$

For capacitor C , if $v = V_m \cos(\omega t + \phi)$
 $\Rightarrow i = -\omega C V_m \cos(\omega t + \phi - 90^\circ)$
 In phasor domain, $I = j\omega C V \Rightarrow V = (1/j\omega C)I$

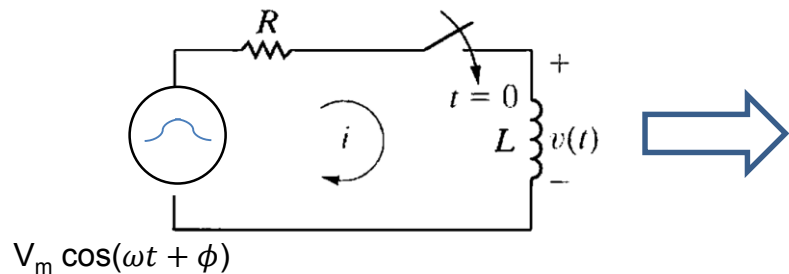
Generic form
 $V = ZI$

Circuit Element	Impedance	(Imaginary part) Reactance
Resistor	R	—
Inductor	$j\omega L$	ωL
Capacitor	$j(-1/\omega C)$	$-1/\omega C$

Practical Example: **Transmission lines**

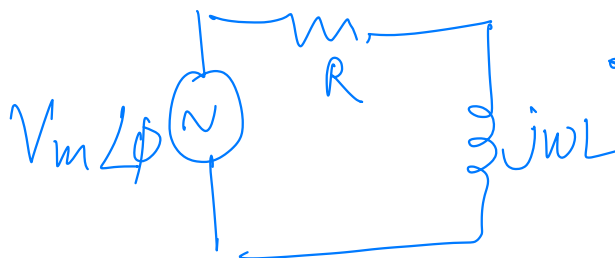


Analyzing AC response in Phasor's Domain



Steady-state analysis

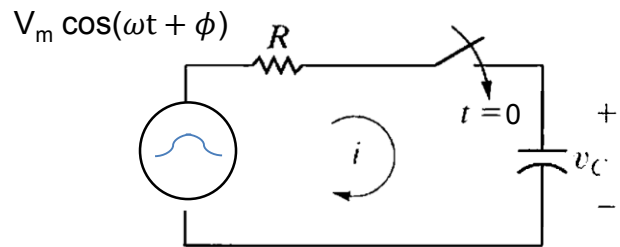
Circuit in phasor domain with reactance



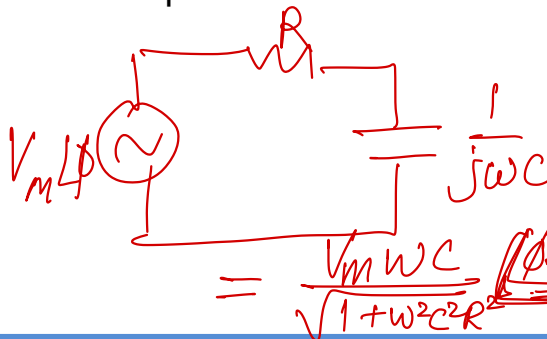
$$I = \frac{V_s}{R + j\omega L} = \frac{V_m L \phi}{\sqrt{R^2 + (\omega L)^2} \angle \tan^{-1}(\frac{\omega L}{R})}$$

$$= \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \angle \phi - \tan^{-1}(\frac{\omega L}{R})$$

$$i(t) = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \cos(\omega t + \phi - \tan^{-1}(\frac{\omega L}{R}))$$



Circuit in phasor domain with reactance



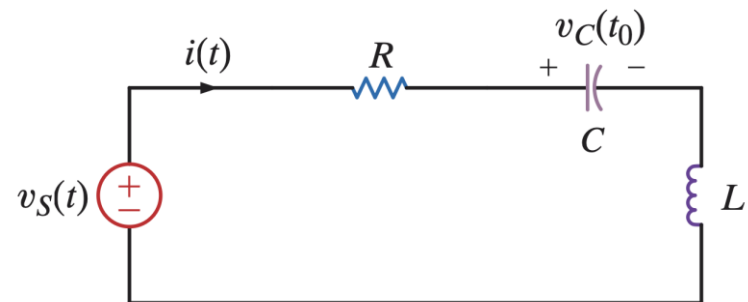
$$\Rightarrow I = \frac{V_m L \phi}{R + \frac{1}{j\omega C}} = \frac{V_m \cdot \omega C L \phi}{\omega C R - j}$$

$$= \frac{V_m \cdot \omega C L \phi}{\sqrt{1 + \omega^2 C^2 R^2}} \angle -\tan^{-1} \frac{1}{\omega C R}$$

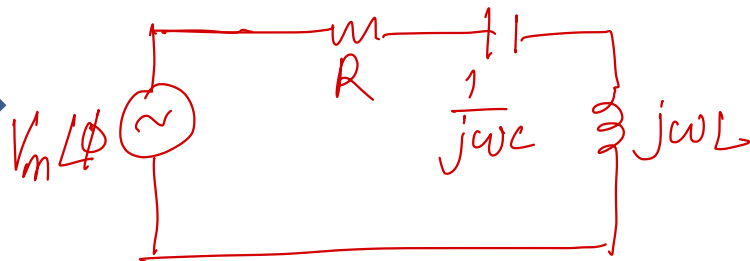
$$= \frac{V_m \omega C}{\sqrt{1 + \omega^2 C^2 R^2}} \angle \phi + \tan^{-1} \frac{1}{\omega C R}$$

→ After converting the circuit into the phasor domain, we can use all our analysis methods (nodal/mesh etc.) discussed earlier for linear circuits.

AC response of circuits combining R, L and C



Circuit in phasor domain with impedance



$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} = \frac{dv_S}{dt}$$

$$I = \frac{V_m \angle \phi}{R + j\omega L + \frac{1}{j\omega C}} = \frac{V_m \angle \phi}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

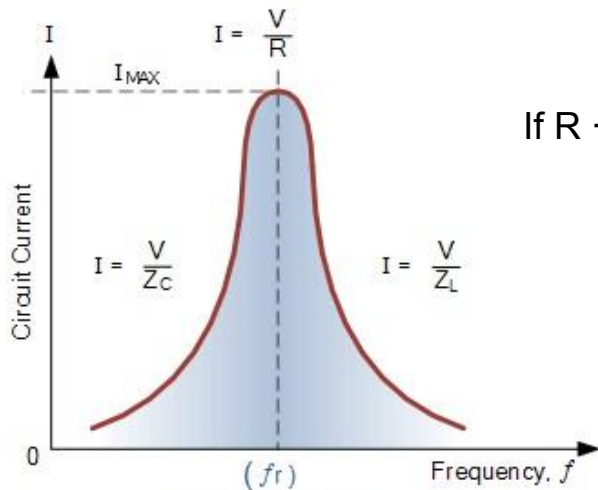
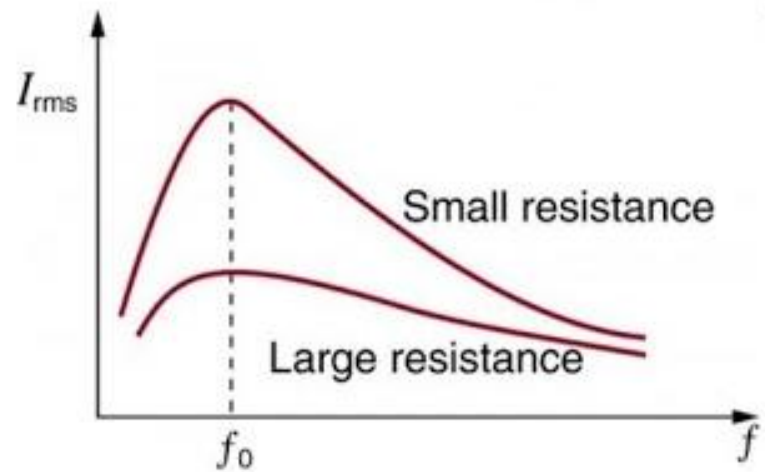
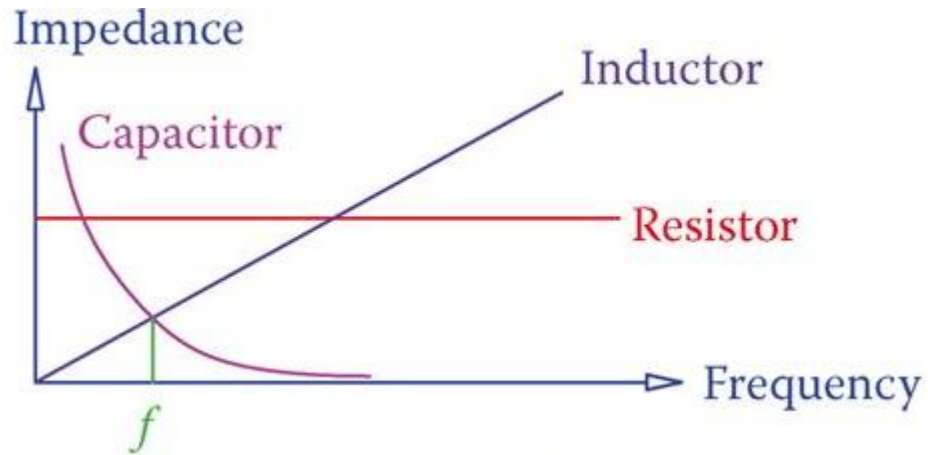
$$= \frac{V_m \angle \phi}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \angle \tan^{-1} \left(\frac{\omega^2 LC - 1}{\omega CR} \right)$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

What if $\omega L = \frac{1}{\omega C}$

$$\Rightarrow I = \frac{V_m \angle \phi}{R \angle 0} = \frac{V_m \angle \phi}{R} \quad (\text{highest } I \text{ at } \omega_0)$$

Visualizing in the frequency domain



If $R \rightarrow 0$, What happens to the system?



Application:
Radio Tuners

Analogy to Mechanical Systems

